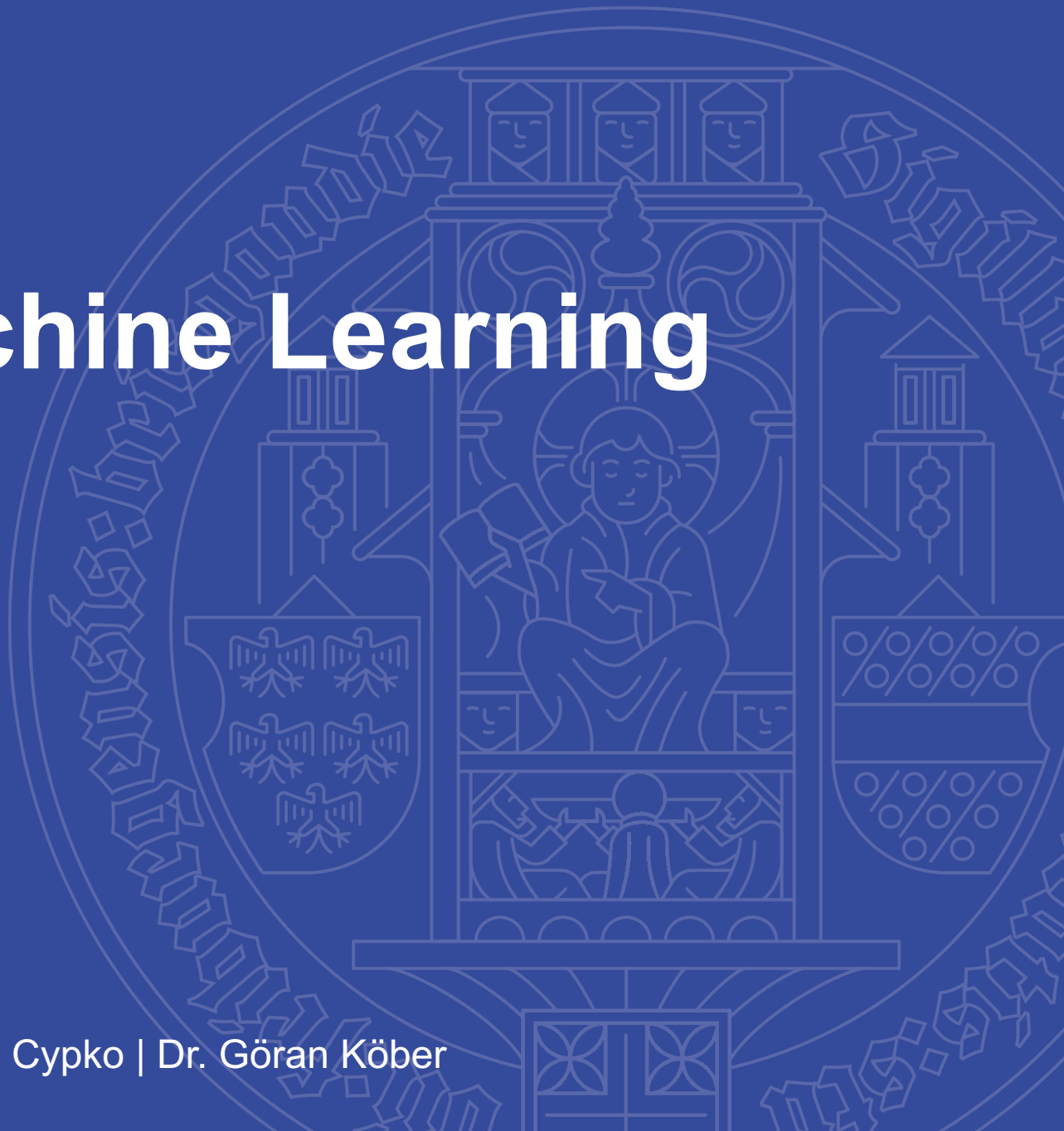
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Intelligent Embedded Systems Lab

Organisation & Machine Learning Refreshers

Exercise 1
22.10.2026
Aditya Kumar, MSc.



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Organisation



How are the exercises structured?

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Thursdays 16–18, online via Zoom. Study weeks have no session: you work on your own until the deadline.

Topic 0 **Refreshers**	**E1** 22.10 Refreshers				
Topic 1 **Medical data**	**E2** 29.10 Introduction	**E3** 05.11 Study week	**E4** 12.11 Study week	**Deadline** **18.11.** 23:59	**E5** 19.11 Discussion & pitch
Topic 2 **Temporal data modelling**	**E6** 26.11 Introduction	**E7** 03.12 Study week	**E8** 10.12 Study week	**Deadline** **16.12.** 23:59	**E9** 17.12 Discussion & pitch
Topic 3 **Bayesian networks**	**E10** 21.01 Introduction	**E11** 28.01 Study week	**Deadline** **16.12.** 23:59	**E12** 04.02 Discussion	

How does one exercise topic work?

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Self-study blocks: experiment with real medical data and modelling tools (e.g. LSTMs, Bayesian networks) at your own pace.

1 Introduction
session

We introduce the exercise, data and tools. Your questions.

2 Study weeks
self-study

Work through guide, task notebook and question sheet.

3 Submission
before the deadline

Task notebook + answer sheet via the hub.

4 Solutions
on the hub

Uploaded after the deadline, before the discussion.

5 Discussion
session

Exercises, solutions, and your own findings beyond the exercise.

optional

required for SL

optional

Pitch (optional): present an additional finding in about 5 minutes. How the discussion runs depends on your interest: come with questions.

Studienleistung (SL)

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What counts

- Topics 1 and 2: the task notebook and the answer sheet, submitted on the hub before the deadline.
- Most exercises are Python notebooks (Jupyter).

How to pass

- Each task and sheet must pass: PASS (≥ 70 %) or PARTIAL PASS (≥ 40 % of the points).
- The notebook must run from top to bottom: we re-run it to grade it.

No SL if …

- a submission is missing or late (submit() closes at the deadline),
- the task or sheet does not pass, or the code does not run.

Register in the ILIAS course by 29.10.2026

Only registered students can take part in the exercises and earn the SL. Requests after the deadline will not be accepted.

The sessions are optional; the submissions are obligatory.

Automatic checks and submission

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1 **Set your ID**

Run the first cells and set `STUDENT_ID = "5123456ak"`: your splits and numbers are personal.

2 **Solve and check**

Fill a TODO, then run `check("1.1")`: ✅ PASS · 🟡 PARTIAL · ❌ FAIL · 📝 recorded (graded after submission).

3 **See your grade**

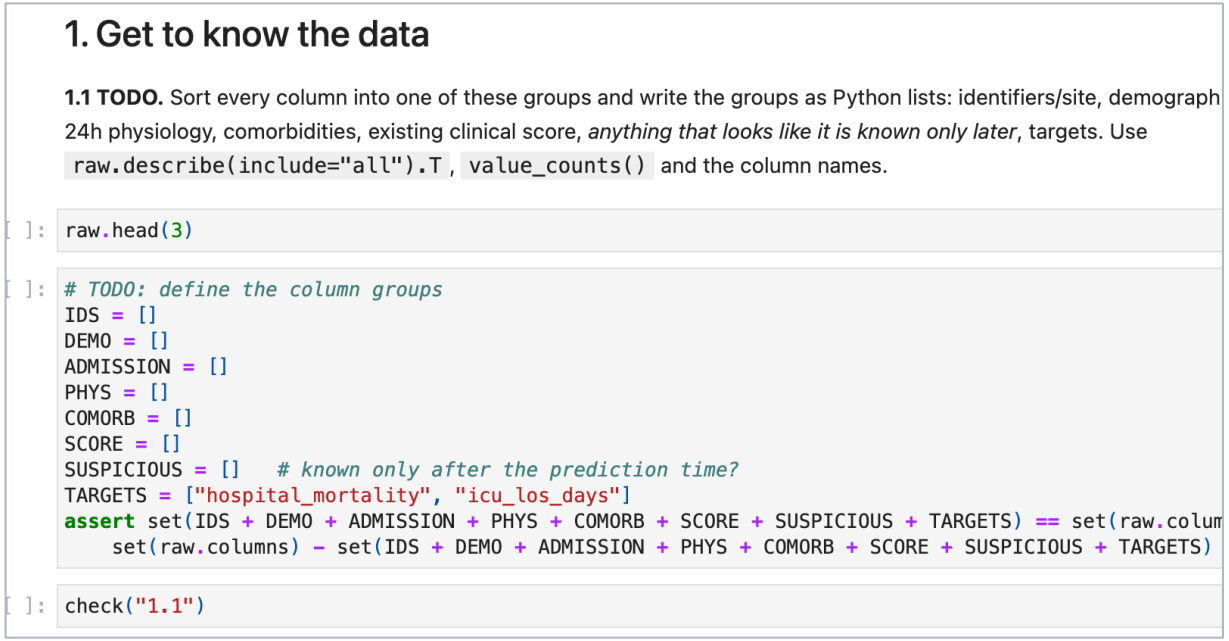
`check_all()` at the end shows your points: PASS ≥ 70 %, PARTIAL PASS ≥ 40 % (both count).

4 **Submit**

Save (Ctrl+S), then run `submit("task1")`. Answer sheet: `submit_file("AnswerSheet_Sheet1_<id>.xlsx")`.

5 **Resubmit any time**

The latest submission before the deadline counts. Afterwards submit() closes and the notebook becomes read-only.



1. Get to know the data

1.1 TODO. Sort every column into one of these groups and write the groups as Python lists: identifiers/site, demograph 24h physiology, comorbidities, existing clinical score, *anything that looks like it is known only later*, targets. Use `raw.describe(include="all").T`, `value_counts()` and the column names.

```
[ ]: raw.head(3)

[ ]: # TODO: define the column groups
IDS = []
DEMO = []
ADMISSION = []
PHYS = []
COMORB = []
SCORE = []
SUSPICIOUS = []   # known only after the prediction time?
TARGETS = ["hospital_mortality", "icu_los_days"]
assert set(IDS + DEMO + ADMISSION + PHYS + COMORB + SCORE + SUSPICIOUS + TARGETS) == set(raw.colum
    set(raw.columns) - set(IDS + DEMO + ADMISSION + PHYS + COMORB + SCORE + SUSPICIOUS + TARGETS)

[ ]: check("1.1")
```

Task 1 notebook: TODO cells, each followed by a check cell

Question sheets and answer sheets

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Part B: Practice questions

Solutions: see *Solution Sheet 1*.

Q1 Rare outcomes and threshold-free metrics *[Multiple correct]*

A mortality model has AUROC 0.85 on a test set where 1% of patients die. Which statements are **necessarily** true?

- **(A)** A random classifier has an expected AUPRC of about 0.01 on this test set.
- **(B)** A classifier that predicts "survives" for everyone has an accuracy of 99%.
- **(C)** If the model is applied to a population with 10% mortality and the score distributions *within* survivors and non-survivors stay the same, the AUROC stays 0.85.
- **(D)** In the same scenario the AUPRC stays the same.
- **(E)** At a fixed threshold, PPV does not depend on prevalence.

answer box	enter	format
Q1	Statements that are necessarily true	letters, e.g. A,C

Q2 Alert burden on a ward *[Numerical]*

An ICU-transfer warning score has sensitivity 0.75 and specificity 0.90. On a ward, 200 patients are scored per day and 5% of them deteriorate.

- **(a)** How many alerts per day are expected, and how many of them are true (1 decimal)? What is the PPV (3 decimals)?
- **(b)** The rapid-response nurse can handle at most 15 alerts per day. Keeping sensitivity at 0.75, what specificity is needed (4 decimals)?

answer box	enter	format
Q2.a1	Expected alerts per day	1 decimal
Q2.a2	True alerts per day	1 decimal
Q2.a3	PPV	3 decimals
Q2.b	Required specificity	4 decimals

id	answer
NAME	
VARIANT	base
Q1	
Q2.a1	
Q2.a2	
Q2.a3	
Q2.b	
Q3	
Q4.a_low	
Q4.a_high	
Q4.b	
Q4.c	
Q5	
Q6.a	
Q6.b	

Answer sheet (.xlsx): one row per answer box

Personal sheet

Your sheet is in `my-sheets/` on the hub.
Numbers differ between students.

Part A and Part B

Worked examples first, then practice questions (multiple choice, numerical, short answer).

Answer boxes

Each question names its boxes (e.g. `Q2.a1`) and the format: enter exactly that in the answer sheet, then `submit_file(…)`.

Use of AI tools

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Welcome as a tutor

- Explaining concepts, error messages, how a pandas or PyTorch function works.
- Start every chat with the official tutor prompt: AI_TUTOR_PROMPT.md in your course folder.
- It makes the assistant give hints and ask questions, not final answers or complete code.

Not allowed

- Pasting eICU or AI-READI data, tables, outputs or screenshots with values into online AI tools (data licences, see Primer C).
- Letting AI produce your answers: your numbers are personal (STUDENT_ID), so copied results do not fit.

Declare it: every task notebook ends with an **`AI_USE`** cell. Write briefly what you used AI for, or "none". Honest declarations do not affect the grade.

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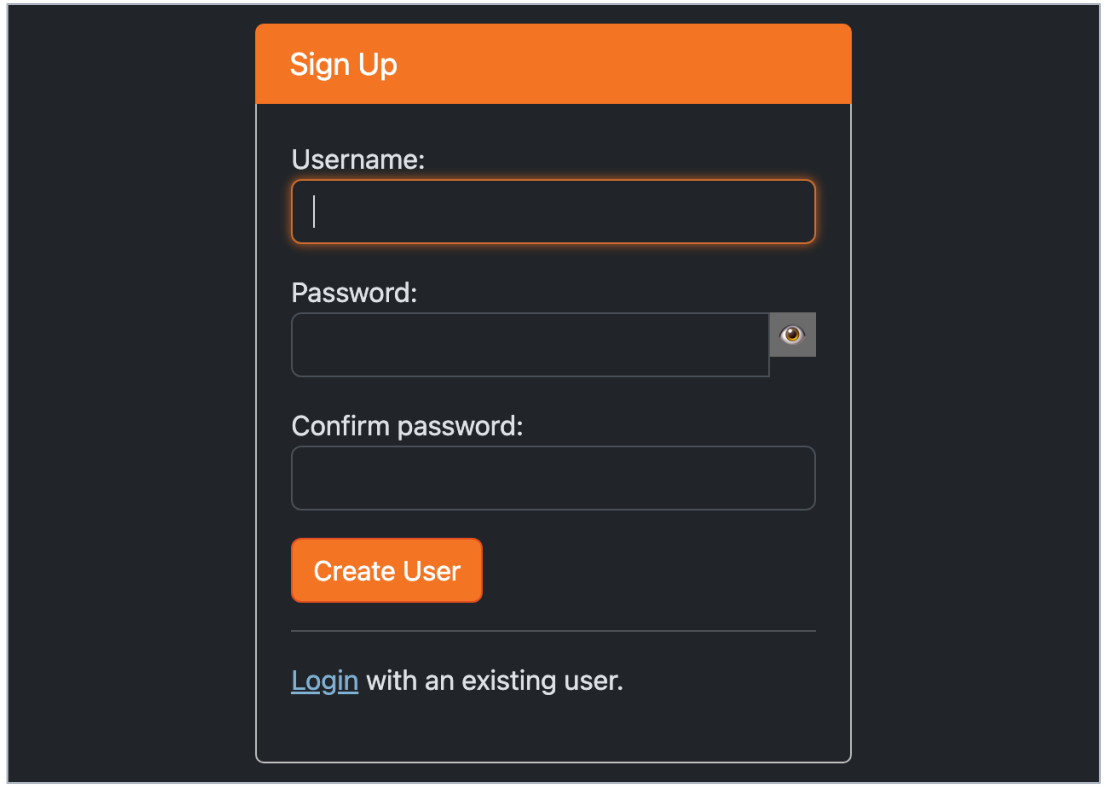
Getting started



The course hub

eghi.bexome.com/digitalhealth26

No installation	JupyterLab in your browser, with data and packages ready.
User name	matriculation number + initials, lower case, e.g. `5123456ak` Use the same ID as STUDENT_ID in the notebooks.
Password	At least 10 characters. Forgotten? Write to me; it cannot be reset by yourself.
Approval	I activate your account after checking it against the ILIAS list.
Deadline	Sign up on the hub and register in ILIAS by 29.10.2026.



Sign-up page of the hub

Live demo: from sign-up to the first notebook

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1 **Open the hub**

eghi.bexome.com/digitalhealth26

2 **Sign up**

User name **`5123456ak`** (matriculation number + initials), password ≥ 10 characters

3 **Wait for approval**

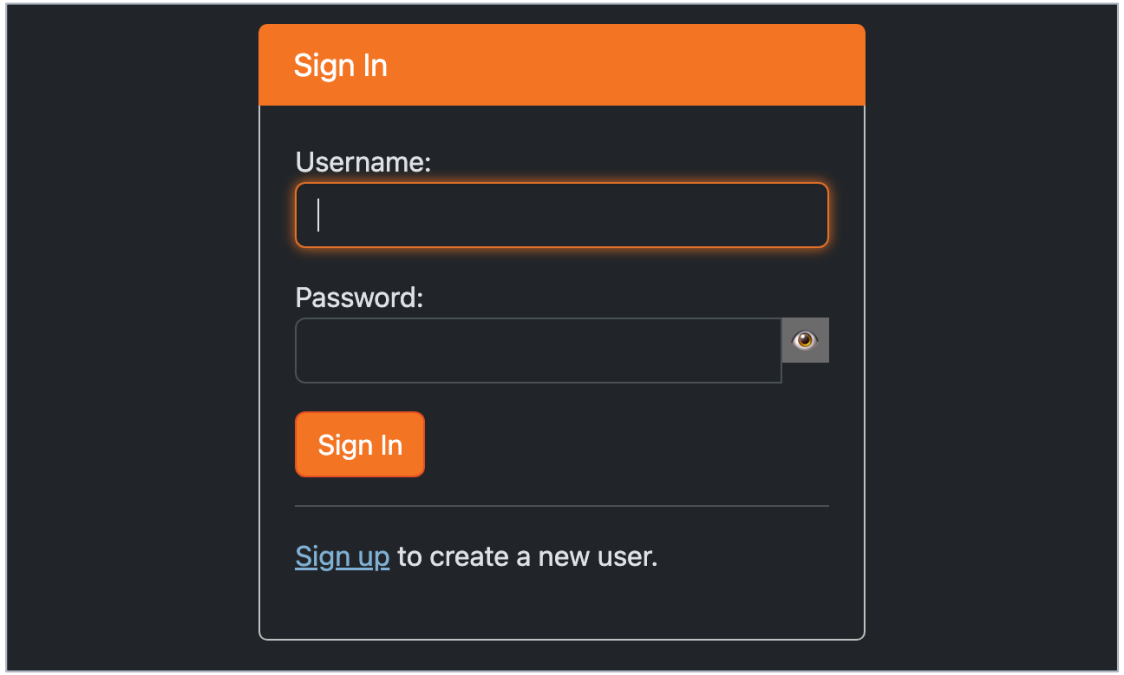
Log-in works once I have activated your account.

4 **Log in**

Your own JupyterLab starts in about 30 s.

5 **Open course/**

Start with `00a_refresher_pandas_clinical.ipynb`.



Servers stop after 1 h without activity; your files are kept. Log in again to continue.

Topic 0: refreshers

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00a

pandas for clinical data

long vs wide, groupby, merges, time zones, reindexing, rolling windows, missing codes

00b

Probability & ML basics

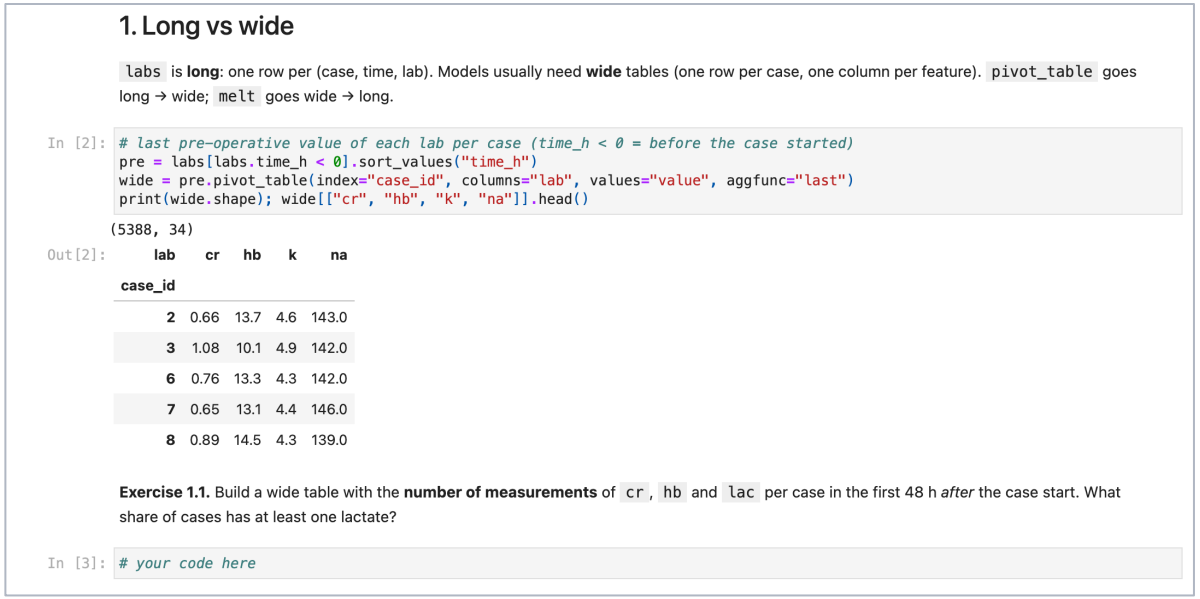
Bayes, odds and logit, confidence intervals, bootstrap, grouped train/test splits

00c

PyTorch & LSTMs

tensors, training loop, LSTM cell by hand, decay, attention, missing-data mechanisms

No submission for topic 0
but finish it before 29.10: the topics build on it.



1. Long vs wide

`labs` is **long**: one row per (case, time, lab). Models usually need **wide** tables (one row per case, one column per feature). `pivot_table` goes long → wide; `melt` goes wide → long.

In [2]:
```
# last pre-operative value of each lab per case (time_h < 0 = before the case started)
pre = labs[labs.time_h < 0].sort_values("time_h")
wide = pre.pivot_table(index="case_id", columns="lab", values="value", aggfunc="last")
print(wide.shape); wide[["cr", "hb", "k", "na"]].head()
```

```
(5388, 34)
```

Out[2]:

lab	cr	hb	k	na
case_id				
2	0.66	13.7	4.6	143.0
3	1.08	10.1	4.9	142.0
6	0.76	13.3	4.3	142.0
7	0.65	13.1	4.4	146.0
8	0.89	14.5	4.3	139.0

Exercise 1.1. Build a wide table with the **number of measurements** of `cr`, `hb` and `lac` per case in the first 48 h *after* the case start. What share of cases has at least one lactate?

In [3]:
```
# your code here
```

Refresher 00a: worked example, then an exercise with a solution to compare

Clinical primers

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Short PDFs with the clinical background for each topic: read the one for a topic before its introduction session.

A

Primer A

Surgery & ICU

Anaesthesia, intensive care, scores and outcomes behind the VitalDB and eICU data.

on the hub from 29.10

C

Primer C

Data governance

Data protection, licences, de-identification: why some data must never leave the hub.

on the hub from 29.10

B

Primer B

Diabetes & wearables

Glucose, CGM metrics and what consumer wearables really measure.

on the hub from 19.11

Location: `course/primers/` on the hub. Primer C applies to every topic that uses restricted data.

Before the next session (29.10)

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- [ ] **Register in ILIAS**
  by 29.10.2026, later requests are not accepted
- [ ] **Sign up on the hub**
  Student_id (e.g., ak1234), lower case; by 29.10.2026
- [ ] **Work through 00a-00c**
  no submission, but you need it for topics 1–3
- [ ] **Join E2 on 29.10, 16:15-17:45**
  Medical data: introduction (Zoom). Guide 1, Task 1, Sheet 1, Primers A and C appear on the hub that day

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Thank you for your attention!

Questions?